

SECURITY INCIDENT REPORT

SOC Analysis Home Lab — Bologo Michael Shumani

1. INCIDENT DETAILS

Incident ID	INC-2026-001
Date & Time	03 May 2026 — 13:32 SAST
Severity	Medium (Rule Level 7–11)
Incident Type	Network Port Scan — Reconnaissance Activity
Affected System	WIN-BG1EIQRHP98 (Windows Server 2022)
Affected IP	172.31.228.160
Source IP	172.31.224.1 (Host Laptop)
Detection Tool	Wazuh SIEM + Sysmon
Analyst	Bologo Michael Shumani
Status	Resolved — Simulated Attack (Lab Environment)

2. EXECUTIVE SUMMARY

On 03 May 2026 at approximately 13:32 SAST, the Wazuh SIEM detected reconnaissance activity targeting the Windows Server 2022 endpoint (WIN-BG1EIQRHP98) within the SOC Analysis Home Lab environment. The activity was identified as an Nmap port scan originating from the analyst's host laptop (172.31.224.1). The scan was initiated intentionally as part of a controlled attack simulation to validate the SOC lab's detection and alerting capabilities.

3. INCIDENT TIMELINE

- 13:30 — Wazuh Agent confirmed active and connected to Wazuh Manager (Ubuntu VM)
- 13:32 — Nmap port scan initiated from host laptop against SOC VM (172.31.228.160)
- 13:32 — Sysmon detected network connection events on the target endpoint
- 13:32 — Wazuh Agent collected Sysmon logs and forwarded to Wazuh Manager
- 13:33 — Wazuh Dashboard alert triggered — Low severity alert count increased from 163 to 164
- 13:35 — Wireshark packet capture confirmed SYN packets from port scan traffic
- 13:40 — Incident documented and classified as simulated attack

4. TECHNICAL ANALYSIS

The Nmap scan was executed using the following command:

```
nmap -sV -p 1-1000 172.31.228.160
```

The scan targeted ports 1 through 1000 on the Windows Server VM. Sysmon (Event ID 3 — Network Connection) captured the connection attempts in real time and forwarded them via the Wazuh Agent to the Wazuh Manager. The Wazuh dashboard reflected the increased alert count within seconds of the scan commencing.

Wireshark packet capture on the vEthernet (Default Switch) adapter confirmed TCP SYN packets originating from the host laptop, consistent with an Nmap SYN scan pattern. Over 51,000 packets were captured during the analysis window.

5. TOOLS USED

- Sysmon v15.20 — Endpoint telemetry and event logging
- Wazuh Agent v4.7.0 — Log collection and forwarding
- Wazuh Manager v4.8.2 — SIEM analysis and alerting
- Wazuh Dashboard — Real-time alert visualisation
- Nmap v7.99 — Attack simulation (port scanning)
- Wireshark v4.6.4 — Network packet capture and analysis
- Hyper-V — Virtual machine host environment

6. FINDINGS & CONCLUSIONS

- The SOC lab successfully detected and alerted on the simulated Nmap port scan within seconds.
- Sysmon event logging was confirmed operational, capturing network connection events in real time.
- The Wazuh Agent established a stable connection to the Wazuh Manager and forwarded logs without interruption.
- Wireshark confirmed the expected SYN packet pattern consistent with a TCP port scan.
- The custom Wazuh detection rule (ID 100002) was successfully created and applied.
- The lab environment demonstrated end-to-end SOC capability: attack simulation, detection, alerting, and analysis.

7. RECOMMENDATIONS

- Implement automated alerting via email or Slack when medium or high severity events are detected.
- Expand Sysmon ruleset to capture additional threat indicators such as process injection and privilege escalation.
- Schedule regular simulated attack scenarios to test detection coverage.
- Integrate MITRE ATT&CK framework mapping for all custom detection rules.
- Implement network segmentation between VMs using separate virtual switches per security zone.

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